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Effects of resistance training with and without post-exercise aerobic activity on strength and body composition according to individual goals

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Abstract

Background/Objective This study aims to examine the effects of two different training protocols aimed at muscle mass gain and fat reduction on strength development and body composition.

Methods The study began with a total of 36 male participants and was completed with 27 participants (fat reduction group = 14 and muscle mass gain group = 13). Participants (age: 32.74 ± 5.53 years; height: 176.81 ± 6.89 cm; weight: 87.56 ± 12.59 kg) performed a total of 40 training sessions over a period of 14–16 weeks. Both groups engaged in resistance training with different loading and rest structures. Additionally, the fat reduction group performed 25–30 min of post-exercise aerobic activity during each session. Anthropometric data such as height, weight, shoulder circumference, chest circumference, waist circumference, hip circumference, and arm circumference were collected using standardized protocols. Body fat percentage measurements were performed using a bioelectrical impedance analysis device. Muscle strength was measured using specific exercises, including Bench Press, Lat Pulldown, Squat, Military Shoulder Press, Barbell Curl, and Triceps Push Down. Maximum strength for each exercise was recorded at the beginning, middle (after the 8th and 24th workouts), and end (after the 40th workout) of the training intervention.

Results The findings indicate that the fat reduction-focused training program significantly reduced body fat percentage while also supporting some strength gains. The muscle mass gain-focused training program was more effective in promoting strength development and muscle mass gain, while also contributing to a slight reduction in body fat percentage.

Conclusions This study confirms that customization of training protocols according to individual goals is beneficial for optimal results. Future research should incorporate a multidisciplinary approach by including nutrition and recovery strategies to further examine these effects.

Keywords Hypertrophy, Fat reduction, Strength development, Body composition

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Introduction

In contemporary society, the objectives of gaining muscle mass and reducing body fat have become pivotal in the fight against escalating health problems and in enhancing overall quality of life [1, 2]. The rise of sedentary lifestyles and the prevalence of poor dietary habits have contributed to a surge in health issues such as obesity, cardiovascular disease, and type 2 diabetes [3]. These conditions not only reduce life expectancy but also significantly diminish the quality of life [4]. Maintaining a healthy body composition is now recognized as a critical component in mitigating these risks [5]. Achieving and sustaining optimal muscle mass and low body fat percentage can lead to improved metabolic health, better physical performance, and a lower incidence of chronic diseases [6]. Consequently, both fitness enthusiasts and healthcare professionals are dedicating substantial efforts to developing and refining training protocols that effectively promote muscle hypertrophy and fat reduction [7, 8]. These efforts are not only aimed at improving individual health outcomes but also at reducing the broader public health burden associated with chronic illnesses.

Resistance training plays a critical role in achieving the goals of muscle mass gain and fat reduction [3, 9]. It is widely recognized for its ability to induce muscle hypertrophy, enhance metabolic rate, and improve overall body composition. The physiological benefits of resistance training include increased muscle strength and endurance, improved insulin sensitivity, and elevated resting metabolic rate, all of which contribute to more efficient fat burning and muscle growth [10, 11]. Recent evidence underscores these benefits across a variety of male populations. For instance, resistance training has been shown to significantly reduce body fat while increasing lean mass in healthy adults [9] and to improve glycaemic control and body composition more effectively than aerobic exercise in individuals with normal-weight type 2 diabetes [10]. Similarly, long-term resistance training improves strength and mitigates fat accumulation even in clinical populations such as prostate cancer patients undergoing androgen deprivation therapy [11]. In trained individuals, both load and repetition progression strategies yield comparable improvements in strength and muscle size [12]. These findings emphasize that resistance training is not only effective but also versatile in eliciting favorable adaptations across a wide demographic. Furthermore, resistance training can help mitigate the loss of muscle mass associated with aging, thereby promoting better functional capacity and quality of life in older adults [12, 13]. Given these extensive benefits, incorporating resistance exercises into fitness regimens is essential for individuals aiming to optimize their physical health and performance. The variety of resistance training methods available allows for tailored programs that can address

specific fitness goals, making it a versatile and indispensable component of comprehensive health and fitness strategies.

In resistance training, the manipulation of various variables such as intensity, volume, frequency, and rest intervals play a crucial role in determining the effectiveness of a training program [1, 13, 14]. Each of these variables can significantly influence the physiological adaptations that occur as a result of training. For instance, higher intensity and volume are often associated with greater muscle hypertrophy and strength gains, while the frequency of training sessions can impact both muscle recovery and growth [15–17]. Similarly, appropriate rest intervals between sets and exercises are essential for optimizing performance and ensuring adequate recovery, thereby maximizing training benefits [18]. The interaction of these variables creates a complex training environment where small adjustments can lead to markedly different outcomes. Understanding how to fine-tune these parameters is essential for designing effective training protocols tailored to individual goals, whether the focus is on maximizing muscle mass, reducing body fat, or improving overall physical performance. This nuanced approach allows for personalized training regimens that can more effectively meet the diverse needs and objectives of different individuals.

The evaluation of different resistance training strategies is essential for identifying effective pathways toward specific fitness goals [19, 20]. While previous research often focuses on isolated variables or controlled populations, there remains a need for comprehensive studies conducted under real-life conditions. This study addresses this gap by investigating the outcomes of two distinct, goal-oriented training approaches: one focused on muscle mass gain through resistance training alone, and the other combining resistance training with post-exercise aerobic activity for fat reduction. Importantly, participants self-selected into these groups based on their personal goals, allowing for an ecologically valid assessment of training outcomes that aligns with individual preferences and motivations. The fat reduction group utilized supersets—a method known for elevating metabolic rate—and included 25–30 min of post-resistance aerobic activity. In contrast, the muscle mass gain group followed a traditional hypertrophy-oriented resistance training regimen. Participants were instructed to maintain their usual dietary habits throughout the intervention to isolate training effects. Across 40 sessions, this study assessed changes in strength and body composition, offering practical insights into the real-world effectiveness of goal-specific training pathways.

Materials and methods

Experimental approach to the problem

This study employed a quasi-experimental, goal-oriented design to examine the outcomes of two distinct training strategies on strength development, muscle hypertrophy, and fat reduction. Rather than being randomly assigned, participants voluntarily selected the training program that aligned with their personal fitness goals—either muscle mass gain or fat reduction—thus enhancing the ecological validity of the study. While training frequency and duration were standardized across both groups, the structure and content of the programs varied based on the intended outcome. The fat reduction group followed a concurrent training approach that incorporated supersets and short rest intervals, along with post-resistance low-intensity aerobic activity to maximize metabolic stress. The muscle mass gain group followed a traditional resistance training protocol focused on hypertrophy, with higher loads and longer rest periods.

Participants completed a total of 40 training sessions over 14–16 weeks, starting with a shared anatomical adaptation phase (first 8 sessions) designed to prepare all individuals for their respective interventions. The dependent variables included strength development, body fat percentage, and anthropometric measures such as weight, circumference (shoulder, chest, waist, hip, arm), and estimated fat-free mass. To control for dietary influence, participants were instructed to maintain their habitual eating patterns throughout the study. This design allowed for a practical, goal-based evaluation of the two training modalities under real-life conditions, offering insights into how different exercise approaches may produce meaningful physiological adaptations when aligned with individual goals.

Participants

The study initially included 36 male participants, who were divided into two groups of 18, targeting either fat reduction or muscle mass gain. However, due to the dropout of 9 participants at various stages of the study, it was completed with 27 participants (age: 32.74 ± 5.53 years; height: 176.81 ± 6.89 cm; weight: 87.56 ± 12.59 kg): 14 in the fat reduction group and 13 in the muscle mass gain group. Participants were pre-screened to ensure comparable baseline fitness profiles. Only individuals aged 18–40 years, who had not engaged in regular physical training in the past six months and reported similar self-rated fitness levels, were included. Although group assignment was based on participant preference, this screening process minimized baseline disparities. The groups were carefully selected to ensure similarity in age and initial fitness level. Participants were recruited through announcements in various gyms, university campuses, and social media platforms. Initial applications

were screened via online surveys and health assessments, and suitable candidates were selected. Selected participants were thoroughly informed about the study's purpose, duration, and requirements, and their consent was obtained. The choice of training focus was made by the participants themselves, with most of those with higher body fat opting for the fat-burning training program.

Inclusion criteria for the study required participants to be between 18 and 40 years old, generally healthy without chronic illnesses, and to have not engaged in regular exercise for at least the past six months to ensure a similar baseline fitness level. Additionally, participants needed to be physically and temporally capable of adhering to the designated training program throughout the study period, with a requirement to attend at least 90% of the sessions, meaning they had to complete a minimum of 36 training sessions. Exclusion criteria included individuals with heart disease, metabolic disorders, or other significant health issues, those using any performance-affecting drugs or hormone supplements, regular strength or resistance trainers, those who consistently used dietary supplements prior to and during the study, individuals with specific dietary restrictions due to health conditions, and those with serious psychological or emotional issues requiring medication.

Training intervention

Anatomical adaptation and basic strength gain common training program

This study included a program of full-body exercises implemented during the first 8 training sessions, aimed at achieving anatomical adaptation and basic strength gain before proceeding to the main 14–16-week (32 sessions) training program designed for specific goals. The program was adjusted to be suitable for beginners in terms of intensity and set/rep schemes.

Each training session targeted all major muscle groups and consisted of 8 exercises, each performed for 3 sets. The weight for each exercise was approximately 60–70% of the participant's one-repetition maximum (1RM), with a tempo of 2 s for both the concentric (lifting) and eccentric (lowering) phases to ensure controlled movement and maximize muscle engagement. For most exercises, including the Smith Machine Bench Press (chest), Machine Shoulder Press (shoulders), Machine Lat Pull-down (back-lats), Machine Leg Press (quadriceps), Leg Curl (hamstrings), Machine Biceps Curl (biceps), and Push Down (triceps), participants completed 12–15 repetitions per set. The Abdominal Machine (abs) was performed for 15–20 repetitions per set to strengthen the abdominal muscles and enhance core stability.

Each set was followed by a 90-second rest period. Training sessions began with a 10–15-minute cardiovascular warm-up, followed by light dynamic warm-up

exercises. Proper form and technique were emphasized during the lifts to ensure safety and effectiveness. After each session, participants performed a 10-minute cool-down and stretching routine. The total workout duration, including the warm-up (15 min), cool-down (10 min), approximately 350–400 repetitions across 24 sets (10–15 min), and rest periods ($24 \times 1.5 \text{ min} = 36 \text{ min}$), was around 65–75 min. This structured program was designed to help participants adapt to the training regimen and develop basic strength, preparing them for the more intensive subsequent training phases.

Fat reduction priority training program

This program was designed to target fat loss by accelerating metabolism and improving body composition (Table 1). Training sessions were conducted 2–3 times per week. In the fat reduction-focused training program, the rest period between supersets was kept between 45 and 60 s. This approach was chosen to keep the heart rate elevated, thereby aiding in increased calorie burning. Different exercises allowed for a 90-second rest period to ensure readiness for the next set without excessive recovery. Training sessions were spaced 48–72 h apart (2 or 3 days) to provide sufficient time for muscle recovery. Exercises were performed with a controlled tempo, typically

1 s for lifting and 3 s for lowering, to reduce injury risk and enhance muscle engagement through increased eccentric loading and motor unit recruitment [21, 22]. The estimated training intensity corresponded to approximately 65–75% of 1RM, adjusted based on individual capacity and progression.

Low-intensity, long-duration cardiovascular exercises were added at the end of each training session (e.g., 25–30 min of running or cycling) to boost fat loss. Participants were encouraged to drink adequate water throughout the day to support body functions and the fat-burning process. However, they were asked to maintain their usual diet to isolate the effects of the training intervention. Total workout duration, including warm-up (10 min), cool-down (10 min), post-workout aerobic exercises (30 min), and the main training session (45–50 min), was approximately 90 min. Some training periods involved repeating the previous weeks' programs to consolidate progress. This approach ensured continuity for muscle development and fat loss, helping reinforce gains. The intensity and volume of workouts were adjusted based on individual progress and capacity.

Table 1 Fat reduction training program details

Session No.	Training Details and Set/Rep Information
1	Super Set 1: Bench Press & Lat Pull Down (5×10 -8-6-6-6) Super Set 2: Incline Bench Press & Bent-Over Barbell Row (4×10 -10-8-8) Super Set 3: Flat-Bench Dumbbell Fly & Reverse Grip Lat Pull Down (3×15)
2	Super Set 1: Dumbbell Shoulder Press & Hanging Leg Raise (5×10 -8-6/15–20) Super Set 2: Dumbbell Lateral Raise & Weighted Decline Crunch (4×10 -10-8/15–20) Super Set 3: Bent-Over Lateral Raise & Dumbbell Shrug (3×15 /12–15)
3	Super Set 1: Incline Bench Dumbbell Curl & V-Bar Push Down (5×12 -10-8) Super Set 2: Standing Barbell Curl & Lying Ez-Bar French Press (4×8 -10) Super Set 3: Preacher Curl & Triceps Dip Machine (3×15)
4	Barbell Squat (5×10 -8-6) Super Set 1: Single Leg Press & Romanian Deadlift (4×10 -12/12–15) Super Set 2: Leg Extension & Standing Calf Raise (3×15 /12–15)
5–8	Repeat sessions 1–4
9	Deadlift (5×10 -8-6) Super Set 1: Flat-Bench Dumbbell Press & Front Lat Pull-Down (4×8 -10) Super Set 2: Decline Press & T-Bar Row (3×12 -15)
10	Super Set 1: Barbell Shoulder Press & Hanging Leg Raise (5×10 -8-6/15–20) Super Set 2: Bent-Over Lateral Raise & Weighted Decline Crunch (4×15 /15–20) Super Set 3: Plate Front Raise & Dumbbell Shrug (3×15 /12–15)
11	Super Set 1: Single-Arm Preacher Curl & Single-Arm Skull-Crusher (5×10 -8-6/8–10) Super Set 2: Standing Barbell Curl & Seated Overhead Dumbbell Extension (4×8 -10) Super Set 3: Incline-Bench Dumbbell Curl & Triceps Dip Machine (3×15)
12	Barbell Squat (5×10 -8-6) Super Set 1: Dumbbell Walking Lunge & Seated Calf Raise (4×10 -12/12–15) Super Set 2: Leg Extension & Lying Leg Curl (3×12 -15)
13–16	Repeat sessions 9–12
17–24	Repeat sessions 1–4
25–32	Repeat sessions 9–12

Note: Unless otherwise stated, all "Squat" exercises refer to barbell back squat, and all "Bench Press" exercises refer to flat barbell bench press

Muscle mass and strength gain priority training program

The focus of strength training workouts is to provide maximum stimulus to the muscles and promote muscle fiber growth (Table 2). These workouts were conducted with higher weights, typically 75–85% of the participant's 1RM, and longer rest periods of 2–3 min to ensure complete muscle recovery and maximize performance in subsequent sets. Each exercise was performed with a controlled tempo, usually 2 s for the concentric (lifting) phase and 2–3 s for the eccentric (lowering) phase. This controlled tempo helps maintain tension in the muscles throughout the entire range of motion, reducing the risk of injury and ensuring effective muscle engagement. During the exercises, it was emphasized that joints should not be fully locked out at the end of the movement. This

approach prevents excessive stress on the joints and keeps the tension focused on the muscles, promoting continuous muscle engagement and growth.

The total workout duration was structured to be approximately 90 min, including various components: a 10-minute dynamic warm-up to prepare the body for intense physical activity, leading to a total of 150–200 repetitions which typically lasted 15–20 min, rest periods of 2–3 min between sets (25–30 sets) to allow for adequate muscle recovery, totaling around 50–60 min, and a 10-minute cool-down of stretching and low-intensity exercises to help reduce muscle stiffness and promote recovery. This structured program was meticulously designed to help participants adapt to the training regimen, enhance their muscle mass, and develop basic to

Table 2 Muscle mass and strength gain training program details

Session No.	Training Details and Set/Rep Information
1	Squat: 3 sets x 8 reps, Bench Press: 3 sets x 8 reps, Good Morning: 3 sets x 8 reps T-Bar Row: 3 sets x 8 reps, Barbell Calf Raise: 3 sets x 8 reps, Barbell Curl: 3 sets x 8 reps
2	Bench Press: 3 sets x 8 reps, Squat: 3 sets x 8 reps, Barbell Row: 3 sets x 8 reps Deadlift: 3 sets x 8 reps, Barbell Calf Raise: 3 sets x 8 reps, Barbell Curl: 3 sets x 8 reps
3	Front Barbell Squat: 3 sets x 8 reps, Barbell Incline Bench Press: 3 sets x 8 reps, Split Squat: 3 sets x 8 reps Pull Up: 3 sets x 8 reps, Barbell Shoulder Press: 3 sets x 8 reps, Triceps Pushdown: 3 sets x 12 reps
4	Squat: 3 sets x 8 reps, Bench Press: 3 sets x 8 reps, Good Morning: 3 sets x 8 reps T-Bar Row: 3 sets x 8 reps, Barbell Calf Raise: 3 sets x 8 reps, Barbell Curl: 3 sets x 8 reps
5–12	Repeat sessions 1–4
13	Squat: 5 sets x 5 reps, Dumbbell Bench Press: 5 sets x 6–8 reps, Facepull: 4 sets x 16 reps One Arm Dumbbell Row: 5 sets x 8–10 reps, Seated Dumbbell Shoulder Press: 5 sets x 10 reps Stable Dumbbell Lunge: 4 sets x 10 reps, Standing Dumbbell Curl: 4 sets x 10 reps, French Press: 4 sets x 10 reps
14	Barbell Bench Press: 5 sets x 5 reps, Pec Dec Fly: 4 sets x 8–12 reps, Rack Pull: 3 sets x 8–12 reps Wide Grip Pull Up: 5 sets x 8–12 reps, Seated Dumbbell Lateral Raise: 4 sets x 8–12 reps Hammer Curl: 8 sets x 8–12 reps, Rope Triceps Extension: 4 sets x 8–12 reps Leg Extension: 4 sets x 6–8 reps, Leg Curl: 4 sets x 6–8 reps
15	Deadlift: 4 sets x 4–6 reps, Incline Dumbbell Bench Press: 4 sets x 8–12 reps Dumbbell Lateral Raise: 5 sets x 8–12 reps, Dumbbell Shrug: 4 sets x 8–12 reps Reverse Pull Up: 4 sets x 8–12 reps, Leg Press: 4 sets x 8–12 reps Barbell Curl: 4 sets x 8–12 reps, Bench Dips: 3 sets x 8–12 reps
16	Barbell Bench Press: 5 sets x 5 reps, Pec Dec Fly: 4 sets x 8–12 reps, Rack Pull: 3 sets x 8–12 reps Wide Grip Pull Up: 5 sets x 8–12 reps, Seated Dumbbell Lateral Raise: 4 sets x 8–12 reps Hammer Curl: 8 sets x 8–12 reps, Rope Triceps Extension: 4 sets x 8–12 reps Leg Extension: 4 sets x 6–8 reps, Leg Curl: 4 sets x 6–8 reps
17–24	Repeat sessions 13–16
25	Back Squat: 5 sets x 4–6 reps, Sumo Squat: 4 sets x 8–12 reps, Bench Press: 5 sets x 4–6 reps Dumbbell Bench Press: 4 sets x 8–12 reps, Bent Over Reverse Barbell Curl: 4 sets x 6–8 reps Cable Straight Push Down: 4 sets x 6–8 reps, Dips: 4 sets x 6–8 reps
26	Front Squat: 4 sets x 6–8 reps, Dip Sumo Squat: 4 sets x 8–12 reps Overhead Barbell Press: 4 sets x 4–6 reps, Overhead Dumbbell Press: 4 sets x 8–12 reps Incline Bench Press: 4 sets x 8–12 reps, Romanian Barbell Deadlift: 4 sets x 6–8 reps Dumbbell Deadlift: 4 sets x 8–12 reps, Rackpull: 3 sets x 6–8 reps
27	Back Squat: 5 sets x 4–6 reps, Hip Squat: 4 sets x 8–12 reps, Paused Bench Press: 5 sets x 4–6 reps Incline Dumbbell Bench Press: 4 sets x 8–12 reps, Bent Over Reverse Barbell Row: 4 sets x 4–6 reps Seated Long Row: 3 sets x 8–12 reps, Farmers Carry: 4 sets x 20 steps
28	Back Squat: 5 sets x 4–6 reps, Sumo Squat: 4 sets x 8–12 reps, Bench Press: 5 sets x 4–6 reps Dumbbell Bench Press: 4 sets x 8–12 reps, Bent Over Reverse Barbell Curl: 4 sets x 6–8 reps Cable Straight Push Down: 4 sets x 6–8 reps, Dips: 4 sets x 6–8 reps
29–32	Repeat sessions 25–28

Note: Unless otherwise stated, all "Squat" exercises refer to barbell back squat, and all "Bench Press" exercises refer to flat barbell bench press

advanced strength, effectively preparing them for the more intensive subsequent training phases. The intensity and volume of workouts were adjusted based on individual progress and capacity. Some training periods involved repeating the previous weeks' programs to consolidate progress. This approach ensured continuity for muscle development and helping reinforce gains.

Data collection process

In this study, detailed analysis of participants' anthropometric measurements and body composition was conducted, along with assessments of muscle strength through various tests. Anthropometric data, including height, weight, shoulder circumference, chest circumference, waist circumference, hip circumference, and arm circumference, were collected using standardized protocols to provide a comprehensive profile of physical characteristics and morphology. Body composition measurements, such as body fat percentage and lean body mass, were obtained using a bioelectrical impedance analysis (BIA) device. Muscle strength was measured using specific exercises, including Bench Press, Lat Pulldown, Squat, Military Shoulder Press, Barbell Curl, and Triceps Push Down. Maximum strength (estimated 1RM) for each exercise was recorded at the beginning, mid-point (after the 8th and 24th sessions), and end (after the 40th session) of the training intervention.

Anthropometric and body composition measurement methods

Body weight was measured using a body composition analyzer (Tanita Corporation, MC-780, Tokyo, Japan) that utilizes BIA. Participants stepped onto the measurement platform without heavy clothing, shoes, or accessories. This device sends a light electrical current through the body, calculating total body weight based on lean mass and fat mass. Height was measured with a wall-mounted stadiometer (Holtain Ltd., UK) with a precision of ± 0.1 cm. Participants stood with their backs against the wall, heels, buttocks, and shoulders touching the wall, with their heads positioned in the Frankfort horizontal plane. Measurements were taken while participants stood straight and breathed normally. The movable part of the stadiometer was lowered gently to touch the highest point of the head, and the measurement was recorded. Each measurement was repeated twice, and a third measurement was taken if there was more than a 0.1 cm difference between the first two measurements. Body Mass Index (BMI) was calculated by dividing body weight (kg) by the square of height (m) (Formula 1).

Formula 1: $BMI = (\text{Body weight (kg)}) / (\text{Height (m)}^2)$

Shoulder circumference was measured at the widest point of the shoulders while participants stood straight with their arms hanging naturally at their sides. A flexible,

thin tape measure was used to ensure accurate measurement without compressing the skin. The tape measure was wrapped around the upper part of the arms and over the shoulder blades, with measurements taken at the widest point. Each measurement was repeated twice, and a third measurement was taken if there was more than a 0.1 cm difference between the first two measurements.

Chest circumference was measured to assess the width of the thorax. This measurement was taken at the widest point of the chest, typically at the level of the nipples, while participants stood straight with their arms hanging naturally at their sides. The tape measure was positioned horizontally around the chest, snug against the skin but not compressing it. Measurements were taken at the midpoint between maximum inhalation and exhalation. Each measurement was repeated twice, and a third measurement was taken if there was more than a 0.1 cm difference between the first two measurements.

Waist circumference was measured to evaluate body fat distribution, particularly in the abdominal area, which is a key indicator of cardiovascular and metabolic health risks. This measurement was taken at the narrowest point of the waist, usually just above the navel. Participants stood straight with their arms away from their bodies to facilitate accurate placement of the tape measure. The tape measure was snug against the skin but not tight, and measurements were recorded during normal breathing, specifically during exhalation. Each measurement was repeated twice, and a third measurement was taken if there was more than a 0.1 cm difference between the first two measurements.

Hip circumference was measured at the widest point of the hips, typically at the most prominent part of the buttocks. Participants stood straight with their feet together to ensure accurate measurement. The tape measure was wrapped horizontally around the hips, ensuring it was snug but not compressing the skin. Measurements were taken during normal breathing, specifically during exhalation. Each measurement was repeated twice, and a third measurement was taken if there was more than a 0.1 cm difference between the first two measurements.

Arm circumference was measured to assess upper body muscle mass and fat distribution, particularly the development of the biceps and triceps. This measurement was taken at the widest point of the dominant arm, usually near the midpoint of the upper arm, close to the elbow joint. Participants raised their arms to a 45-degree angle and made a fist to naturally contract the biceps. The tape measure was wrapped horizontally around the arm, ensuring it was snug but not compressing the skin. Each measurement was repeated twice, and a third measurement was taken if there was more than a 0.1 cm difference between the first two measurements.

Strength measurement methods

Participants underwent strength assessment tests including Bench Press, Lat Pulldown, Squat, Military Shoulder Press, Barbell Curl, and Triceps Push Down. For these tests, weights that allowed participants to complete 5–10 repetitions were selected. This repetition range is considered ideal for ensuring safety and estimating maximal strength values. The 1RM was estimated using the Brzycki formula [23–25]. This formula provides an accurate estimate of maximal strength based on lower repetition counts. If participants struggled with the selected weight, it was reduced by 3–10%. Conversely, if participants easily surpassed 10 repetitions, the weight was increased similarly. This method enhanced test accuracy and better reflected participants' true strength levels. A complete range of motion was required for a valid repetition, with safety being a priority throughout the test. An observer or trainer was present to ensure correct form and assist during the tests.

Formula 2: 1RM (kg) = (Weight lifted (kg) × 36) / (37 - Number of repetitions)

Bench press test

Primary muscles: Pectoral muscles (chest)

Secondary muscles: Triceps and anterior deltoids (front shoulders)

Participants prepared for the bench press exercise with appropriate warm-up activities, including general body warm-up and specific movements targeting the chest, shoulders, and arms. Following the warm-up, an initial weight of approximately 50% of body weight was selected. The 1RM strength test was performed using the standard flat barbell bench press, with participants lying on a flat bench and using a straight barbell. Participants lay on the bench press apparatus with their feet firmly planted on the ground. The barbell was held at chest level, and participants were instructed to fully lift and lower the barbell. Proper form was continuously monitored, ensuring hands were placed slightly wider than shoulder-width apart on the bar. During the lift, arms were fully extended without locking the elbows, keeping tension on the muscles rather than the joints. The barbell was lowered until it lightly touched the chest without bouncing.

Lat pulldown test

Primary muscles: Latissimus dorsi (major back muscles)

Secondary muscles: Biceps, posterior deltoids (rear shoulders), and rhomboids (middle back muscles)

Participants conducted the lat pulldown test by sitting at the lat pulldown machine (Technogym S.p.A., Cesena, Italy). After warming up with dynamic stretches and light resistance exercises, an initial weight of about 50% of body weight was selected. Participants sat with their feet firmly on the supports and gripped the bar slightly wider

than shoulder-width apart. The bar was pulled down to the chest with a controlled movement, ensuring maximum muscle contraction. The lift involved pulling the shoulders down and back while lowering the bar to the chest, then slowly returning it to the starting position.

Squat test

Primary muscles: Quadriceps (front thigh muscles), gluteus maximus (buttocks muscles)

Secondary muscles: Hamstrings (back thigh muscles), adductors (inner thigh muscles), and soleus (lower calf muscles)

Before the squat test, participants performed a comprehensive warm-up including general body movements and dynamic stretches targeting the lower body. An initial weight of approximately 70% of body weight was selected. The 1RM strength test was conducted using the standard barbell back squat. Participants positioned the barbell on their shoulders and adopted a balanced stance with feet shoulder-width apart and toes slightly pointed outward. They descended slowly, keeping their knees aligned with their toes and lowering their hips as much as possible, ideally until the thighs were parallel to the floor. The ascent involved pushing through the heels to return to the starting position. The back was kept in a neutral position throughout the movement to prevent excessive strain on the spine. An observer or trainer monitored the form and provided support as needed to ensure safety and effectiveness.

Military shoulder press test

Primary muscles: Deltoids (shoulder muscles).

Secondary muscles: Triceps, trapezius (upper back and neck muscles).

Participants warmed up with exercises targeting the shoulders, arms, and upper back to prepare for the military shoulder press. An initial weight of approximately 40% of body weight was selected. Standing with feet shoulder-width apart, participants held the barbell at chest level and pressed it overhead until the arms were fully extended without locking the elbows. The back was kept straight, avoiding excessive arching. The lift emphasized maintaining a stable core and proper form, with the bar held just above the head at the top of the movement. Form was carefully monitored, and corrections were made to ensure proper technique and prevent injury.

Barbell curl test

Primary muscles: Biceps

Secondary muscles: Brachialis and brachioradialis (forearm muscles).

Participants warmed up with general body movements and specific exercises targeting the arms and shoulders. An initial weight of approximately 25% of body weight

was selected. Standing with feet shoulder-width apart and knees slightly bent, participants performed the barbell curl with palms facing upward and elbows close to the body. The barbell was lifted to shoulder height in a controlled manner and then lowered back to the starting position. Full range of motion was ensured, with arms fully extended at the bottom and fully flexed at the top.

Triceps push down test

Primary muscles: Triceps

Secondary muscles: Anconeus (small muscle around the elbow)

Participants warmed up with exercises targeting the upper body, particularly the arms and shoulders, to prepare for the triceps push down. An initial weight of approximately 30% of body weight was selected. Standing at the push down machine with feet shoulder-width apart, participants held the bar attached to the cable pulley (Technogym S.p.A., Cesena, Italy). The bar was pushed down until the arms were fully extended without locking the elbows. The movement was controlled, focusing on keeping the shoulders stationary and using the elbows as the primary joint for the exercise.

Data analysis

The statistical analysis of data obtained from the participants was performed using the Statistical Package for Social Sciences (SPSS) version 25.0 (IBM Corp, Chicago, IL, USA). The normality of the variable distributions was assessed using the Shapiro-Wilk test and histogram plots. Data were presented as mean $\pm$ standard deviation ($\bar{X} \pm SD$). To examine differences in descriptive statistics between groups before the intervention program, an Independent Samples T-Test was used. The effects of the independent variables on the dependent variables, measured before the intervention, and after the 8th, 24th, and 40th sessions of the program, were analyzed using a two-way mixed-design ANOVA (Group = 2 $\times$ Time = 4). The effect size for this test was reported as partial eta squared (η^2), with η^2 values interpreted as small > 0.01 , medium ≥ 0.06 , and large ≥ 0.14 . The assumption of sphericity was evaluated using Mauchly's Test. If the assumption of sphericity was violated and Epsilon (ϵ) was < 0.75 , the Greenhouse-Geisser correction was applied. If ϵ was > 0.75 , the Huynh-Feldt correction was used. Changes in the research groups from before the intervention program to after the 40th training session were summarized using Cohen's d coefficient. The significance of Cohen's d was interpreted based on the following thresholds proposed by Hopkins: trivial (0.00–0.19), small (0.20–0.59), medium (0.60–1.19), large (1.20–1.99), and very large (≥ 2.00). An alpha level of $\alpha < 0.05$ was considered significant for all analyses.

Ethical principles of the study

The study received approval from the Scientific Research and Publication Ethics Committee of Erzurum Technical University (Meeting Number: 16, Decision Number: 04, Date: 28.12.2023) and was conducted in accordance with internationally accepted ethical principles. Before participating in the study, all participants were provided with detailed information about the purpose, procedures, and potential risks of the research. Informed consent was obtained from each participant through a written consent form. Participants' personal information and research data were protected in accordance with confidentiality principles; all data were anonymized and securely encrypted.

Results

Table 3 shows the baseline comparisons between the Muscle Mass Gain Group (MMGG) and the Fat Reduction Group (FRG). There were no statistically significant differences in age, height, or body weight between the groups ($p > 0.05$). Similarly, no significant differences were found in initial strength measures, including bench press, lat pulldown, squat, shoulder press, barbell curl, and triceps push down (all $p > 0.05$). However, there was a significant difference in body fat percentage ($p < 0.001$) and BMI ($p = 0.032$), with the FRG group starting at higher levels. This difference reflects the nature of the study design, as individuals with higher body fat levels naturally chose the fat-reduction-focused training protocol based on their personal fitness goals.

Table 4 presents the results of the pair-wise comparisons for body composition and circumference measurements. In the MMGG, there were no statistically significant differences in body weight and shoulder circumference measurements. However, in the FRG, body weight, BMI, and shoulder circumference significantly decreased after the 8th, 24th, and 40th training sessions compared to the pre-test. Significant reductions were also observed in chest and waist circumference in the FRG, with notable decreases in each subsequent measurement. Hip circumference showed significant reductions in the FRG at each measurement point, while no significant differences were found in arm circumference in either group. In terms of fat-free mass, the MMGG showed a significant increase by the 40th session, indicating improvements in muscle tissue. Conversely, the FRG also exhibited a moderate but significant increase in FFM, despite simultaneous reductions in total body weight and fat percentage. These results suggest that both protocols can contribute to preserving and increasing lean mass, albeit through different training emphases.

Table 5 shows the results of the pair-wise comparisons for strength measurements. Both groups demonstrated significantly higher strength performance after the 40th

Table 3 Descriptive characteristics of participants and comparison of differences between groups

Variable	Group	X±SD	t	p
Age (years)	MMGG	32.77 ± 6.15	0.025	0.980
	FRG	32.71 ± 5.12		
Height (cm)	MMGG	177.38 ± 6.93	0.407	0.687
	FRG	176.28 ± 7.06		
Body Weight (kg)	MMGG	83.61 ± 11.85	1.615	0.119
	FRG	91.21 ± 12.54		
BMI (kg/m ²)	MMGG	26.54 ± 3.15	2.276	0.032
	FRG	29.31 ± 3.17		
Body Fat (%)	MMGG	22.69 ± 3.47	5.568	< 0.001
	FRG	34.71 ± 7.02		
Bench Press (kg)	MMGG	51.60 ± 14.19	0.219	0.828
	FRG	50.43 ± 13.59		
Lat Pulldown (kg)	MMGG	47.65 ± 10.18	0.188	0.853
	FRG	48.33 ± 8.67		
Squat (kg)	MMGG	71.17 ± 18.41	0.262	0.795
	FRG	73.43 ± 25.51		
M. Shoulder Press (kg)	MMGG	29.26 ± 5.43	0.959	0.347
	FRG	31.56 ± 6.94		
Barbell Curl (kg)	MMGG	25.85 ± 4.76	0.317	0.754
	FRG	23.25 ± 4.21		
Triceps Push Down (kg)	MMGG	41.72 ± 10.88	0.129	0.899
	FRG	41.13 ± 12.68		

MMGG: Muscle Mass Gain Group; FRG: Fat Reduction Group; n: Sample size; X: Mean; SD: Standard Deviation

Table 4 Multiple comparison results for body composition and circumference measurements

	MMGG				FRG			
	Baseline	8th session	24th session	40th session	Baseline	8th session	24th session	40th session
	X±SD	X±SD	X±SD	X±SD	X±SD	X±SD	X±SD	X±SD
Weight (kg)	83.62 ± 11.85	83.46 ± 7.07	83.85 ± 6.14	83.54 ± 6.13	91.21 ± 12.54	86.11 ± 9.55 ^a	84.57 ± 8.53 ^a	83.86 ± 7.93 ^{ab}
BMI (kg/m ²)	26.54 ± 3.15	26.55 ± 2.13	26.67 ± 1.83	26.56 ± 1.77	29.31 ± 3.17	27.68 ± 2.18 ^a	27.17 ± 1.73 ^a	26.96 ± 1.62 ^a
Body Fat (%)	22.69 ± 3.47	22.00 ± 2.58	21.69 ± 1.93	20.61 ± 1.89 ^{abc}	34.71 ± 7.02	31.57 ± 6.60 ^a	29.14 ± 5.51 ^{ab}	27.35 ± 5.31 ^{abc}
Fat-Free Mass (kg)	64.45 ± 8.21	65.04 ± 5.43	65.62 ± 4.92	66.31 ± 4.82 ^{abc}	59.29 ± 8.34	58.81 ± 7.59	59.82 ± 6.47	60.85 ± 6.66 ^{abc}
Shoulder Circumference (cm)	111.20 ± 9.63	110.92 ± 7.69	111.55 ± 8.05	111.88 ± 8.16	113.15 ± 9.13	111.04 ± 9.71 ^a	110.05 ± 9.71 ^a	109.32 ± 9.67 ^{ab}
Chest Circumference (cm)	100.79 ± 8.33	101.72 ± 7.85	102.28 ± 7.86	101.98 ± 8.01	105.20 ± 9.43	103.81 ± 7.65	102.52 ± 6.55	101.47 ± 6.21 ^{bc}
Waist Circumference (cm)	90.07 ± 8.98	89.31 ± 8.86	88.00 ± 7.82	87.31 ± 7.59	102.79 ± 10.52	100.26 ± 9.62 ^a	98.11 ± 9.43 ^{ab}	97.30 ± 9.43 ^{abc}
Hip Circumference (cm)	91.50 ± 11.69	90.33 ± 11.12	89.27 ± 9.94	89.35 ± 9.90	102.63 ± 7.26	100.38 ± 7.41 ^a	99.05 ± 7.43 ^{ab}	98.04 ± 6.77 ^{abc}
Arm Circumference (cm)	36.17 ± 3.32	36.36 ± 3.67	36.33 ± 3.35	36.62 ± 3.44	36.52 ± 3.40	35.46 ± 3.82	35.60 ± 3.83	35.54 ± 3.86

MMGG: Muscle mass gain group; FRG: Fat reduction group; X: Mean; SS: Standard deviation

^a: Significantly different from baseline; ^b: Significantly different from 8th session; ^c: Significantly different from 24th session

training session compared to the initial two measurements (pre-test and 8th session). Significant differences emerged after the 8th training session in the Military Shoulder Press, Barbell Curl, and Triceps Push Down tests, while other tests showed significant differences after the 24th session. In the MMGG, strength performance significantly increased across all tests after the

40th session, whereas in the FRG, no significant difference was found between the 24th and 40th sessions for the Military Shoulder Press test.

Figure 1 illustrates the changes in body weight and circumference measurements for both training groups. Significant differences were found over time for body weight, with no significant differences between groups,

Table 5 Multiple comparison results for strength measurements

	MMGG				FRG			
	Baseline	8th session	24th session	40th session	Baseline	8th session	24th session	40th session
	$\bar{X} \pm SD$	$\bar{X} \pm SD$	$\bar{X} \pm SD$	$\bar{X} \pm SD$	$\bar{X} \pm SD$	$\bar{X} \pm SD$	$\bar{X} \pm SD$	$\bar{X} \pm SD$
Bench Press (kg)	51.60 $\pm$ 14.19	59.82 $\pm$ 9.93	64.15 $\pm$ 10.23 ^{ab}	69.08 $\pm$ 8.90 ^{abc}	50.43 $\pm$ 13.59	53.73 $\pm$ 9.01	58.81 $\pm$ 8.81 ^{ab}	65.02 $\pm$ 8.86 ^{abc}
Lat Pulldown (kg)	47.65 $\pm$ 10.18	54.52 $\pm$ 9.83	58.11 $\pm$ 8.91 ^{ab}	63.57 $\pm$ 7.05 ^{abc}	48.33 $\pm$ 8.67	49.32 $\pm$ 11.52	55.41 $\pm$ 9.68 ^{ab}	60.16 $\pm$ 8.62 ^{abc}
Squat (kg)	71.17 $\pm$ 18.41	77.05 $\pm$ 16.93	82.77 $\pm$ 17.18 ^{ab}	95.44 $\pm$ 18.72 ^{abc}	73.43 $\pm$ 25.51	74.45 $\pm$ 20.10	80.48 $\pm$ 16.40 ^{ab}	90.77 $\pm$ 15.98 ^{abc}
M. Shoulder Press (kg)	29.26 $\pm$ 5.43	31.12 $\pm$ 5.29 ^a	34.01 $\pm$ 7.49 ^a	36.90 $\pm$ 7.86 ^{abc}	31.56 $\pm$ 6.94	33.01 $\pm$ 7.26 ^a	34.81 $\pm$ 8.51 ^a	35.94 $\pm$ 7.91 ^{ab}
Barbell Curl (kg)	25.85 $\pm$ 4.76	29.19 $\pm$ 3.99 ^a	32.93 $\pm$ 5.74 ^{ab}	37.45 $\pm$ 5.01 ^{abc}	23.25 $\pm$ 4.21	27.08 $\pm$ 5.41 ^a	31.55 $\pm$ 6.58 ^{ab}	35.64 $\pm$ 6.61 ^{abc}
Triceps Push Down (kg)	41.72 $\pm$ 10.88	46.18 $\pm$ 9.63 ^a	54.38 $\pm$ 9.66 ^{ab}	58.41 $\pm$ 9.63 ^{abc}	41.13 $\pm$ 12.68	45.76 $\pm$ 10.98 ^a	51.03 $\pm$ 10.46 ^{ab}	54.67 $\pm$ 8.68 ^{abc}

MMGG: Muscle mass gain group; FRG: Fat reduction group; $\bar{X}$: Mean; SD: Standard deviation

^a: Significantly different from baseline; ^b: Significantly different from 8th session; ^c: Significantly different from 24th session

although the group $\times$ time interaction was significant. Shoulder circumference showed significant differences over time but not between groups. There were no significant differences in chest circumference over time or between groups, but the group $\times$ time interaction was significant. Both time and group differences were significant for waist and hip circumference measurements, while no significant differences were found for arm circumference.

Figure 2 shows the changes in body fat percentage measurements for both training groups. The analyses indicated significant differences over time, between groups, and in the group $\times$ time interaction. Figure 3 illustrates the changes in strength performance for the two training groups (MMGG and FRG). For the bench press measurements, there was a significant difference over time ($p < 0.001$), but no significant difference between groups ($p = 0.221$), and the group $\times$ time interaction was not significant ($p = 0.604$). Similar results were observed for the lat pulldown, with significant differences over time ($p < 0.001$), no significant differences between groups ($p = 0.434$), and no significant group $\times$ time interaction ($p = 0.226$). Squat measurements showed significant differences over time ($p < 0.001$), but no significant differences between groups ($p = 0.793$) or in the group $\times$ time interaction ($p = 0.301$). The military shoulder press also had significant differences over time ($p < 0.001$), with no significant differences between groups ($p = 0.702$) or in the group $\times$ time interaction ($p = 0.078$). For the barbell curl, there were significant differences over time ($p < 0.001$), but no significant differences between groups ($p = 0.291$) or in the group $\times$ time interaction ($p = 0.887$). Finally, the triceps push down showed significant differences over time ($p < 0.001$), with no significant differences between groups ($p = 0.605$) or in the group $\times$ time interaction ($p = 0.174$).

Figure 4 presents the effect size coefficients for changes in body composition, circumference, and strength measurements over the 40-session intervention period for both training groups. The findings indicate that both muscle mass gain and fat reduction training programs

had a “moderate” effect on body fat percentage. The muscle mass gain program had a “trivial” effect on body weight, whereas the fat reduction program had a “moderate” effect. In terms of body circumference measurements, the muscle mass gain program had a “trivial” effect on arm, hip, chest, and shoulder circumferences, but a “small” effect on waist circumference. Conversely, the fat reduction program had “small” to “moderate” effects on all circumference measurements. For strength measurements, both programs had a “very large” effect on bench press, lat pulldown, barbell curl, and push down performance. The effect size for military shoulder press performance was “large.” Additionally, the muscle mass gain program had a “very large” effect on squat performance, while the fat reduction program had a “large” effect.

Discussion

This study aimed to evaluate the effects of two distinct training protocols, one focused on muscle mass gain and the other on fat reduction, on strength development and body composition. The muscle mass gain program led to significant improvements in strength and muscle mass, though changes in body composition were limited. Conversely, the fat reduction program resulted in significant decreases in body fat percentage and body weight, along with improvements in strength measurements. These findings demonstrate the ability to customize training approaches to target specific goals, producing distinct effects on body composition and strength parameters.

The fat reduction training program utilized high-intensity supersets with short rest periods to maintain high heart rates and increase metabolic stress, promoting high energy expenditure and accelerating fat loss [26, 27]. Importantly, the fat reduction group in this study received a combined intervention (resistance training + aerobic exercise), which mirrors common real-life practice among individuals aiming to improve body composition. Therefore, the body fat reductions observed in this group should be interpreted as the outcome of a

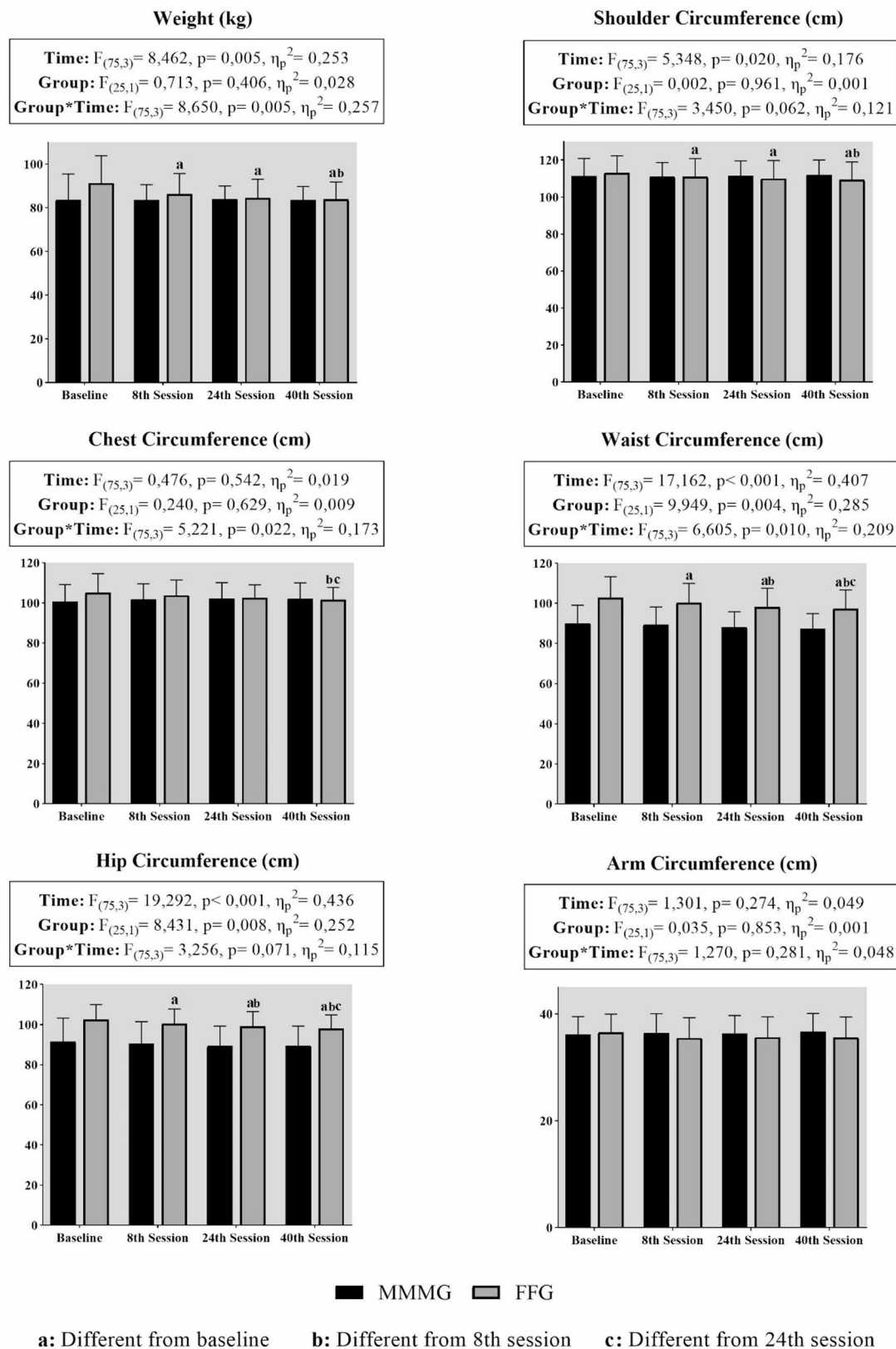


Fig. 1 Changes in Body Weight and Circumference Measurements Depending on Training Interventions

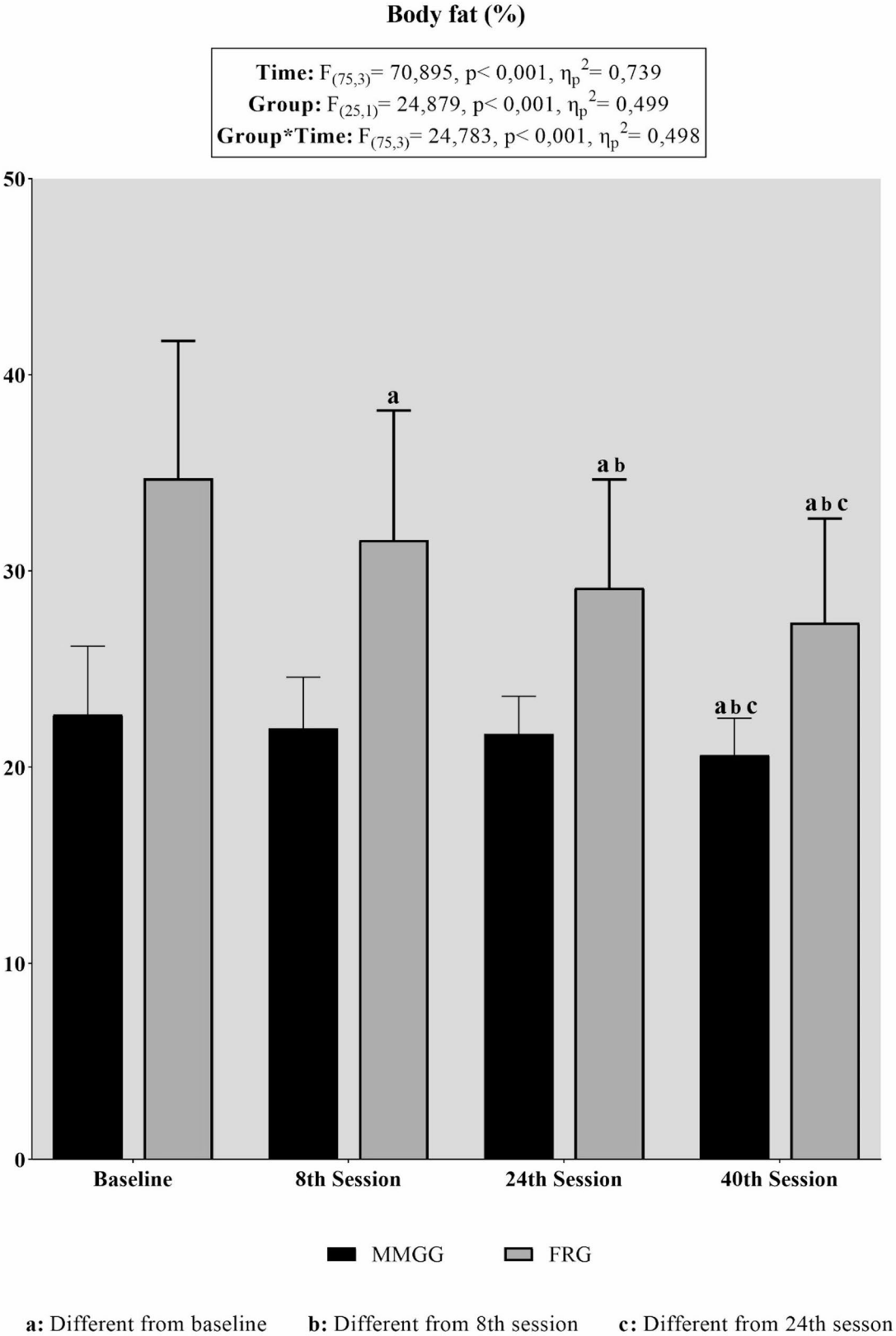
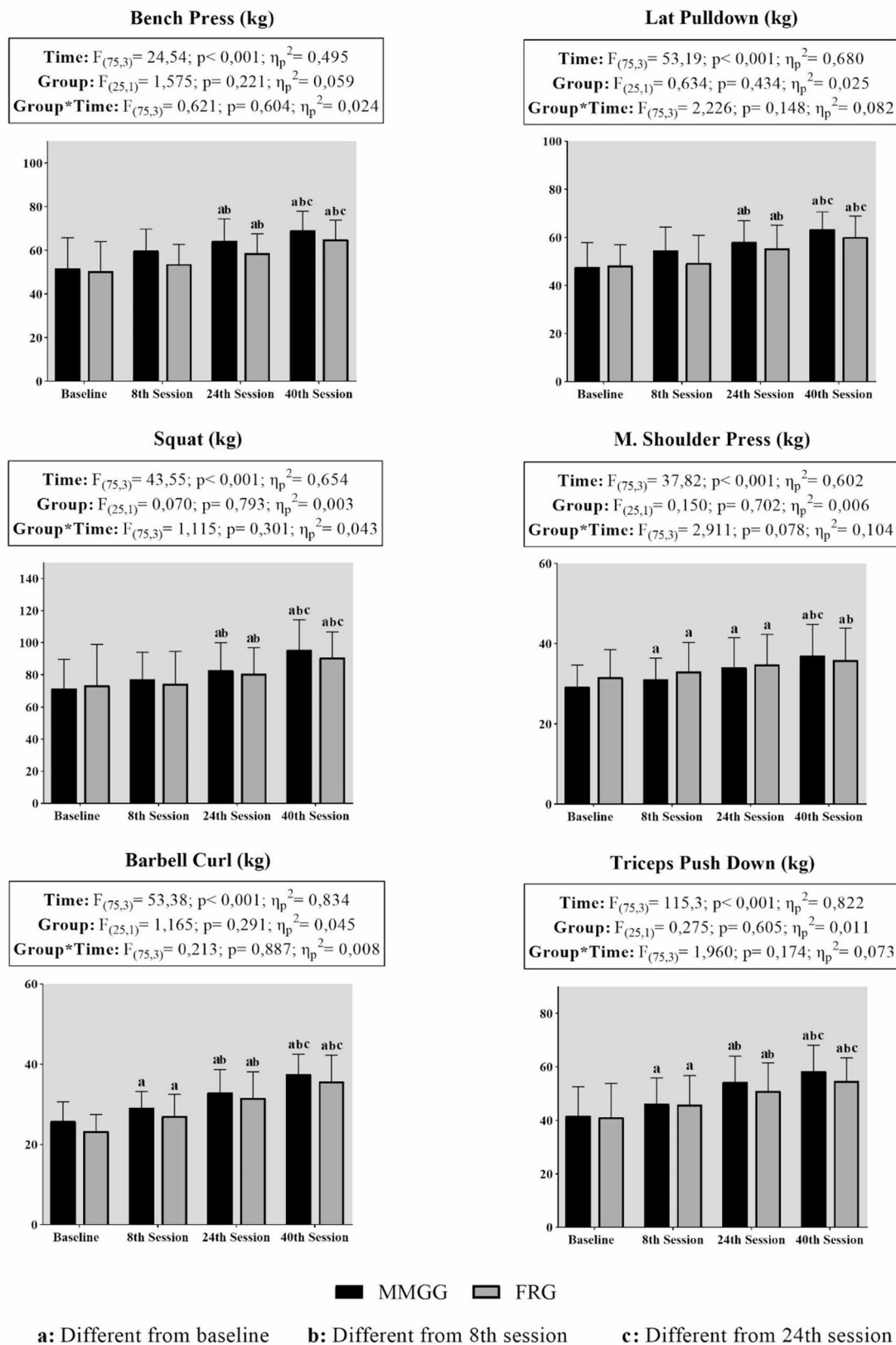


Fig. 2 Changes in Body Fat Percentage due to Training Interventions

**Fig. 3** Changes in Strength Performance Due to Training Interventions

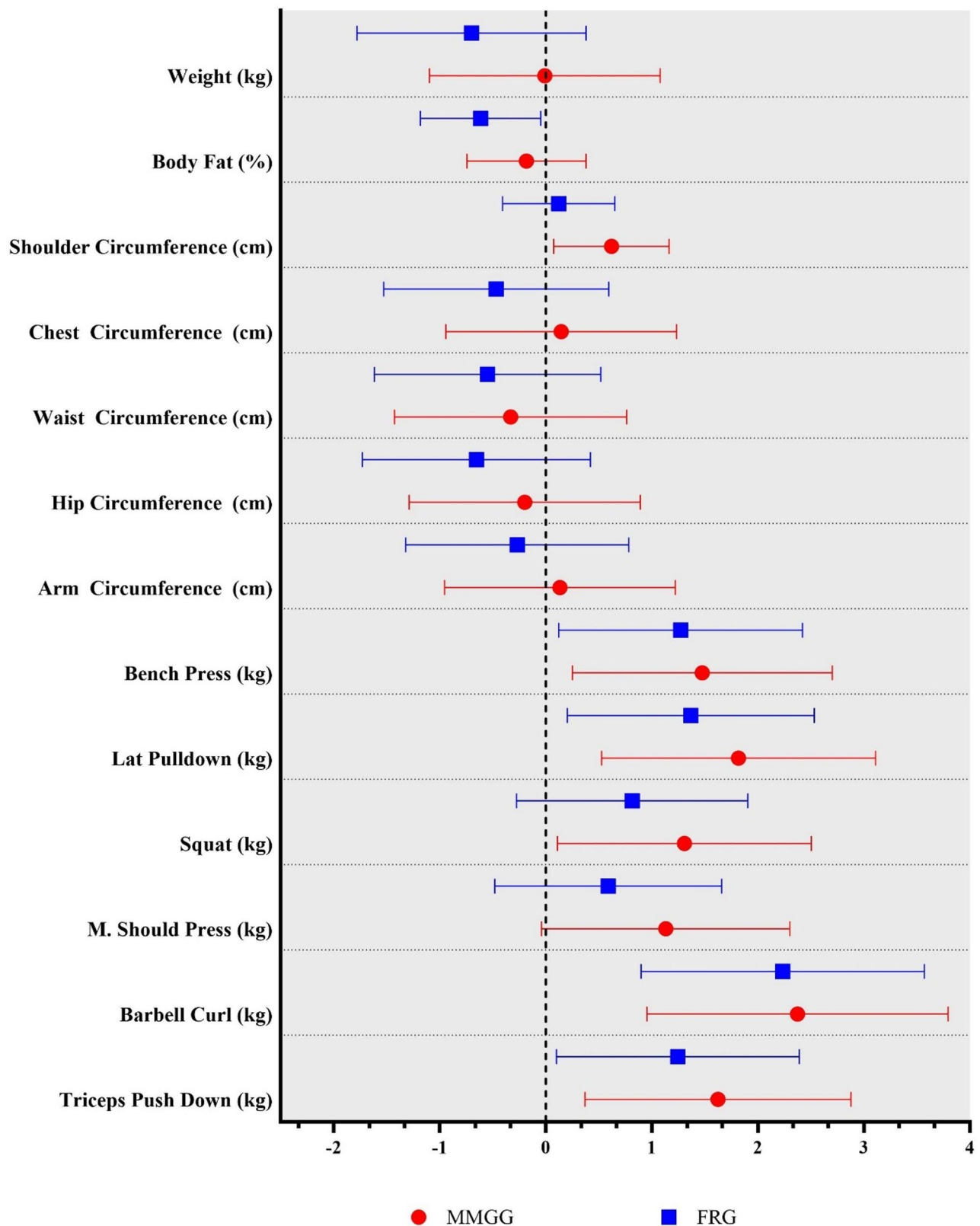


Fig. 4 Practical Importance (Effect Size) Analyses of Changes in the Variables Examined Depending on Training Interventions

combined strategy, rather than resistance training alone. Low-intensity, long-duration cardiovascular exercises further enhanced fat loss [28]. This approach is effective for individuals prioritizing fat loss, as it increases metabolism and caloric expenditure, leading to significant reductions in body fat percentage and body weight. The muscle mass and strength gain training program employed higher weights and longer rest periods, providing maximum stimulus to the muscles and promoting muscle fiber growth [29, 30]. This program focused on fundamental strength movements with progressively increased weights, allowing for full muscle recovery and maximum performance in subsequent sets [31, 32].

Our study observed strength performance increases ranging from 13 to 53% after 40 training sessions over 14–16 weeks. Previous studies have shown that resistance training performed 2–3 times weekly over similar periods can lead to significant improvements. For instance, Rhea et al. (2002) reported changes of 12–56% in bench press and leg press exercises over 12 weeks using different periodization methods [33]. Kibele and Behm (2009) noted an 18% increase in squat performance in the first 4 weeks and a 14% increase from the 4th to the 6th week during a 7-week training period [34]. In our study, while both groups demonstrated notable improvements in strength performance, the MMGG exhibited greater gains across nearly all strength tests compared to the FRG, aligning with the higher resistance intensities and rest intervals tailored for hypertrophy and strength development (Fig. 4).

Schroeder et al. (2019) found that combined aerobic and resistance training led to significant reductions in blood pressure and improvements in cardiorespiratory fitness (CRF) and muscle strength over 8 weeks [35]. Similarly, Ho et al. (2012) reported greater improvements in body weight, total body fat, and CRF with combined training compared to aerobic or resistance training alone in overweight and obese adults over 12 weeks [36]. These findings align with our study, indicating that fat-reduction-focused programs incorporating resistance exercises can lead to significant fat loss and modest strength gains. While group $\times$ time interactions were not statistically significant for strength variables (Fig. 3), effect size analyses revealed a greater magnitude of strength improvement in the MMGG (Fig. 4). This trend suggests that hypertrophy-oriented protocols may offer more pronounced strength adaptations when training volume and intensity are appropriately structured.

The significant strength gains in the fat reduction group can be attributed to the inclusion of high-intensity interval training (HIIT) and supersets, which enhance both cardiovascular endurance and muscle strength [37, 38]. HIIT activates both anaerobic and aerobic energy systems, leading to increased muscle activation and strength

gains [39]. Supersets apply continuous stress on muscles, enhancing both endurance and strength [16]. High-intensity training also increases the secretion of anabolic hormones like growth hormone and testosterone, supporting muscle protein synthesis and strength gains [40, 41]. Additionally, fat reduction programs often include compound exercises targeting large muscle groups, which increase both energy expenditure and muscle strength [17, 42]. Compound movements like squats, deadlifts, and bench presses enhance both metabolic rate and muscle strength, supporting fat loss and strength development simultaneously. Overall, our study's findings align with the existing literature and highlight the importance of customizing training programs to meet specific goals.

Despite its practical design and structured methodology, this study has certain limitations that should be acknowledged. First, although participants were explicitly instructed to maintain their usual dietary routines, no formal dietary assessment was conducted. As such, unmonitored changes in eating behavior—whether intentional or incidental—may have influenced the observed changes in body composition. Second, participants self-selected into groups based on their personal goals rather than being randomly assigned, which may have introduced selection bias and limited internal validity. However, this approach was intentional to enhance the ecological validity of the study and reflect real-world training environments. Third, while training intensity and structure were carefully designed and controlled through estimated %1RM and rest intervals, objective physiological indicators of intensity (e.g., heart rate, lactate concentration, or RPE) and caloric expenditure were not measured. This limits the ability to quantify internal load and energy output precisely. Finally, individual adherence and external lifestyle factors (e.g., sleep, stress, incidental activity) were not monitored and could have affected the outcomes. On the other hand, a major strength of this study lies in its goal-oriented design, which offers practical relevance for fitness professionals and recreational athletes. By applying differentiated training protocols based on individual goals in a semi-controlled setting, this study provides valuable insights into realistic adaptations in strength and body composition. Future research should include direct measurements of physiological intensity and caloric expenditure to provide a more comprehensive understanding of training responses across diverse protocols.

Conclusions

In conclusion, the fat reduction program effectively decreased body fat and body weight while improving strength, whereas the muscle mass gain program enhanced strength and muscle mass, with some

reduction in body fat. However, our study had several limitations, including a relatively small sample size and a lack of control over participants' dietary intake, which may have influenced the results. Additionally, the self-selection of training programs by participants might have introduced bias. Notably, participants in the fat reduction group had significantly higher baseline body fat percentages, which may have contributed to the greater magnitude of fat loss observed in that group. While these differences reflect real-world decision-making and individual training goals, they may limit the direct comparability between the groups. Despite these limitations, the study offers valuable insights into the practical outcomes of goal-oriented training approaches. Future research should address these limitations by incorporating larger sample sizes, controlling for dietary intake, objectively quantifying training intensity and energy expenditure, and randomly assigning participants to training programs to validate and extend our findings.

Practical applications

In practical terms, the fat reduction program can be recommended for athletes and individuals aiming to reduce body fat and improve cardiovascular fitness. The incorporation of high-intensity exercises and short rest periods helps maintain an elevated heart rate and increase metabolic stress, which are critical for fat loss. Additionally, the inclusion of low-intensity, long-duration cardiovascular exercises at the end of training sessions can further enhance caloric expenditure. The muscle mass gain program, focusing on high weights and longer rest periods, effectively promoted muscle hypertrophy and strength gains. This program is ideal for individuals aiming to increase muscle mass and strength. Emphasizing fundamental strength movements with progressively heavier weights and allowing sufficient rest for full muscle recovery ensures maximum performance and muscle growth.

These findings underline the importance of customizing training programs to match individual goals. For athletes and coaches, selecting and designing training protocols that align with specific objectives—whether it be fat loss or muscle gain—can optimize outcomes. It is also important to recognize that individuals with higher baseline body fat may experience more pronounced reductions, particularly when programs integrate both resistance and aerobic elements. For fat loss, combining resistance training methods such as supersets and short rest intervals with post-exercise aerobic activity is advisable, as this integrated approach enhances both metabolic stress and overall energy expenditure. Conversely, for those focused on muscle mass gain, resistance training with higher loads and adequate recovery appears more beneficial. Furthermore, integrating nutrition and

recovery strategies into training programs is crucial for achieving desired results. A multidisciplinary approach that includes tailored dietary plans and adequate recovery periods can significantly enhance the effectiveness of training protocols. Future research should explore the long-term effects of various training protocols to further refine and optimize training strategies. By understanding the prolonged impact of these protocols, sports scientists and practitioners can develop more effective and sustainable training programs that cater to diverse fitness goals.

Abbreviations

MMGG	Muscle Mass Gain Group
FRG	Fat Reduction Group
1RM	One-repetition maximum
BMI	Body Mass Index
HIIT	High-intensity interval training
ηp^2	Partial eta squared

Acknowledgements

This study was derived from a master's thesis. We extend our gratitude to all participants involved in this study for their valuable contributions.

Author contributions

Authors contributed to the concept and design (SU and UA), acquisition of the data (SU and UA), analysis (SU) and interpretation (all authors), drafting and revision (SÖ and SU), final approval (all authors) and agreement to be accountable (all authors). All authors contributed equally to the manuscript and read and approved the final version of the manuscript.

Funding

This study is not funded.

Data availability

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

Declarations

Ethics approval and consent to participate

The study received approval from the Scientific Research and Publication Ethics Committee of Erzurum Technical University (Meeting Number: 16, Decision Number: 04, Date: 28.12.2023) and was conducted in accordance with internationally accepted ethical principles. Before participating in the study, all participants were provided with detailed information about the purpose, procedures, and potential risks of the research. Informed consent was obtained from each participant through a written consent form. Participants' personal information and research data were protected in accordance with confidentiality principles; all data were anonymized and securely encrypted. The study was conducted in accordance with the ethical principles of the Declaration of Helsinki.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 24 October 2024 / Accepted: 3 July 2025

Published online: 28 October 2025

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